

Summary

MS in Computer Science student and software engineer building scalable **AI/ML, cloud, and customer-facing systems**. Experienced in **Python, Java, C#, and TypeScript**, with strong foundations in algorithms, distributed services, REST APIs, **Microsoft Azure, Microsoft Power Platform, and AWS**. Comfortable owning products from requirements and architecture through deployment, validation, and iteration.

Experience

Engineering Intern (Product Development & QA), Dometic May 2026 – Present

- Built a **Next.js AI-powered analytics dashboard**, cutting laboratory-data comparison, processing, and analysis time by **70%**.
- Deployed the dashboard on **Azure App Service** with **Azure Load Balancer**; used **Azure Blob Storage** for storage and **IBM AS/400 and Microsoft Dynamics 365 connectors** to ingest data for unified comparison and analytics.
- Built and trained an **ML forecasting model** on historical data to predict future warranty costs with **under 8% error**, improving inventory and resource planning.
- Automated **employee and project progress tracking** with **Microsoft Power Automate**, streamlining status updates, task follow-ups, and project-management reporting.

Research Assistant (Software/ML Engineering), Illinois Institute of Technology Jan 2026 – Present

- Trained and evaluated **Llama 3.1 and Qwen2.5** alongside **Kronos**, a financial time-series foundation model, and custom CryptexLLM on cryptocurrency data, achieving **58% test directional accuracy**.
- Automated **data preprocessing**, experiment execution, validation, and result aggregation through reusable Python tooling, improving reproducibility and reducing manual effort across repeated runs.
- Ran concurrent large-model training on an **8-GPU NVIDIA Tesla V100 cluster**, distributing and load-balancing workloads across devices to accelerate experiments and improve hardware utilization.

Teaching Assistant, Illinois Institute of Technology Sep 2025 – Jan 2026

- Supported 40+ students through labs, grading, and office hours, applying consistent evaluation criteria and clear technical feedback.
- Reviewed programming assignments for correctness, debugging approach, **data structures**, documentation quality, and problem-solving clarity.

Software Engineer, BarodaWeb Jun 2024 – Dec 2024

- Built, deployed, and hosted **full-stack web features** with React, Node.js, Express, and MongoDB, translating customer requirements into solutions for data capture, configuration, and operational workflows.
- Developed a **camera-integrated computer vision advertising screen** using a **CNN-based age-and-gender estimation model** to analyze viewers in real time and display personalized advertisements.

Software Engineer Intern, VMC Mar 2024 – May 2024

- Modernized a **legacy C#/.NET ticketing platform**, redesigning customer workflows to reduce required clicks by **30%** and improve the user experience.
- Implemented **load balancing and performance optimizations**, improving peak-traffic stability by **10%** and strengthening service reliability.

Projects

CryptexLLM Python, PyTorch, Transformers

- Built a **scalable time-series forecasting pipeline** using OHLCV data, patch-based tokenization, RevIN normalization, and a frozen LLM reprogramming head.
- Validated behavior with walk-forward evaluation, tracking MAE, RMSE, directional accuracy, and risk-adjusted returns under realistic assumptions.

E-Motel React, Node.js, Express, MongoDB, Socket.IO, AWS, Vercel

- Built and deployed a **full-stack management web app** on **AWS** with MongoDB Atlas, Express, React, Node.js, JWT auth, and real-time **Socket.IO** updates for booking and staff workflows.
- Integrated an **AI-assisted chatbot** to automate check-in and streamline operations through real-time updates and automatic task assignment.

Credit Card Fraud Detection Using Transformer Python, PyTorch, scikit-learn

- Built a **Transformer-based detection model** on 284,807 transactions (0.17% fraud), achieving **ROC-AUC 0.9643, Recall 0.8667, and Accuracy 0.9929**.
- Improved performance with a **Feature-Gated FT-Transformer** using Conv1D and sigmoid token gating, boosting **ROC-AUC to 0.9827** and reducing test loss from **0.3360 to 0.2758**.

Skills

Languages: Java, Python, C++, C#, TypeScript, JavaScript, SQL, Go (familiar)

CS Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Distributed Systems, System Design, REST APIs, Concurrency

Cloud & Backend: Microsoft Azure, Azure App Service, Azure Load Balancer, Azure Blob Storage, Azure Data Lake Storage, AWS, Amazon SageMaker, Google Cloud, .NET, Docker, Node.js, Express.js, MongoDB, SQL & NoSQL, GraphQL, Socket.IO, microservices, Microsoft Dynamics 365, IBM AS/400

Frontend: React, Next.js, HTML/CSS, Tailwind CSS

ML & GenAI: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Hugging Face Transformers, Llama 3.1, Qwen2.5, Kronos, time-series forecasting, computer vision, CNNs, distributed/multi-GPU training, LLM evaluation, OpenAI/Claude API, prompt engineering

Tools & Workflow: Microsoft Power BI, Microsoft Power Automate, Git, CI/CD, Linux, VS Code, Postman, debugging, automation

Education

Illinois Institute of Technology Chicago, IL
M.S. in Computer Science Expected Dec 2026

Sardar Vallabhbhai Institute of Technology Anand, India
B.S. in Computer Science May 2024

Hackathons & Awards

MACIA Hackathon – Winner Blockchain, Payment Processing

- Designed, built, and presented a **blockchain-based payment processing prototype** for secure transaction handling and transparent verification, winning the hackathon.

Certifications & Training

- **AWS Cloud Practitioner Essentials (Cloud 101)** – cloud concepts, core AWS services, security, architecture, and pricing fundamentals.
- **Google Cloud (GCR) Student Facilitator** – led peer study sessions and supported students through Google Cloud hands-on labs and learning paths.

Publication

Het Patel, et al. *CryptexLLM*

Research manuscript in preparation on an LLM-powered time-series forecasting pipeline using OHLCV market data, RevIN normalization, and robustness evaluation across multiple training windows.